

AAMER ABDUL

✉ ar.aamer@gmail.com

🌐 linkedin.com/in/aamer-ar/

🐙 github.com/Aamer98

🎓 [Google Scholar](#)

RESEARCH INTERESTS

Vision Language Models, Robustness, Generalization, Explainability, Offline RL, and their applications to Climate and Healthcare

EDUCATION

Mila Quebec — ETS Montréal

2022 – 2024

Research MSc in Computer Science

GPA: 3.91/4.30

Focus: Machine learning and ML robustness. Advisor: Prof. Samira Ebrahimi Kahou

Birla Institute of Technology and Science, Pilani

2017 – 2021

Bachelor of Engineering in Electronics and Communication Engineering

GPA: 8.67/10.00

RESEARCH EXPERIENCE

Parameter Efficient Fine-Tuning of Atmospheric Foundation Models — Mila, Quebec

Jan 2024 – present

Supervisor: Samira E. Kahou, Karthik Mukkavilli (IBM), Lester Mackey (Microsoft)

- Working on effective and efficient fine-tuning regimes to adapt atmospheric ViT-based foundation models to wildfire modeling tasks. Under review at **AGU 2024**.
- Studied and implemented multiple parameter efficient fine-tuning methods such as LoRA, BitFit and SSF on foundation models with multi-modal remote sensing data.

Uncertainty Measures to Mitigate Model Unfairness in Healthcare — Mila, Quebec

Jan 2024 – present

Supervisor: Samira E. Kahou, Bianca Dumitrascu (Columbia University)

- Exploring the use of uncertainty measures to detect and mitigate biased and unfair model behavior on decision making tasks on healthcare.
- Studied and implemented several uncertainty methods such as MC dropout, bayesian neural networks and deep ensembles.

Offline Reinforcement Learning for Medical Decision Making — Mila, Quebec

Sept 2022 – Dec 2023

Supervisor: Samira E. Kahou, Vincent Michalski (University of Montreal)

- Proposed the Medical Decision Transformer (MeDT), a goal conditioned reinforcement learning framework for interpretable and interactive medical decision making. Spotlight at **NeurIPS 2023** workshop.
- Detailed performance validation using off-policy evaluation methods such as weighted importance sampling, fitted Q-evaluation, weighted doubly-robust and model-based algorithms.
- MeDT outperforms or is competitive with existing offline RL algorithms across multiple evaluation metrics.

Survey on Transformers in Reinforcement Learning — Mila, Quebec

Sept 2022 – Dec 2023

Supervisor: Samira E. Kahou, Simon Prince (University of Bath)

- The survey covers the application of transformers in representation learning, transition and reward function modeling, and policy optimization within reinforcement learning.

Uncertainty Measures for Improved Concept Bottleneck Models — Mila, Quebec

May 2022 – Sept 2022

Supervisor: Samira E. Kahou, Ivaxi Sheth (CISPA), Mohammad Havaei (Google)

- Devised an uncertainty based strategy, SIUL, to allow for more effective human-model interaction with concept bottleneck models. Accepted at **NeurIPS 2022** workshop.
- SIUL makes concept bottleneck models more robust to concept leakage, distribution shifts and adversarial attacks.

Impact of Normalization Layers on Cross Domain Few-Shot Transfer — Mila, Quebec

July 2021 – April 2022

Supervisor: Samira E. Kahou, Eugene Belilovsky (Mila), Mohammad Havaei (Google)

- Devised Feature Normalization that improves few-shot generalization performance up to **6%** on shifted domains. Accepted at **CVPR 2022**.
- Feature Normalization improves robustness of few shot transfer to distribution shifts with negligible computational overhead.

PROFESSIONAL EXPERIENCE

Mila, Quebec

July 2021 – April 2022

ML Research Intern. Supervisor: Samira E. Kahou

- Explored and devised normalization schemes to improve performance of few-shot methods to extreme distribution shifts.
- Worked on improving the robustness of explainable concept bottleneck models to distribution shifts and adversarial attacks.

University of Toronto

Dec 2020 – June 2021

ML Research Intern. Supervisor: Chi-Guhn Lee

- Studied and implemented adversarial domain adaptation methods for object detection in data-sparse settings.

Techrobotic FZCO, Dubai

June 2017 – Sept 2017

Mechatronics Engineering Intern

- Programmed interactive applications for touch-screen kiosks. Technologies used: C++ , Arduino.

SELECTED PUBLICATIONS

Efficient Fine-tuning of Atmospheric Foundation Models for Wildfire Modeling

Under review AGU 2024

Aamer Abdul Rahman, Lester Mackey, Karthik Mukkavilli, Samira Ebrahimi Kahou

Empowering Clinicians with Medical Decision Transformers: Paper Code

Preprint: Under review TMLR

Spotlight at NeurIPS 2023 Workshop on Goal-Conditioned RL

Aamer Abdul Rahman, Pranav Agarwal, Rita Noumeir, Philippe Jouvét, Vincent Michalski, Samira Ebrahimi Kahou

Transformers in reinforcement learning: a survey: Paper

Preprint: Under review ACM Computing Surveys

Pranav Agarwal, Aamer Abdul Rahman, Pierre-Luc St-Charles, Simon JD Prince, Samira Ebrahimi Kahou

Learning from uncertain concepts via test time interventions: Paper

NeurIPS 2022 Workshop on Trustworthy and Socially Responsible ML

Ivaxi Sheth, Aamer Abdul Rahman, Laya Rafiee Sevyeri, Mohammad Havaei, Samira Ebrahimi Kahou

Pitfalls of Conditional Batch Normalization for Contextual Multi-Modal Learning: Paper

NeurIPS 2022 ICBINB Workshop

Ivaxi Sheth, Aamer Abdul Rahman, Mohammad Havaei, Samira Ebrahimi Kahou

Revisiting Learnable Affines for Batch Norm in Few-Shot Transfer Learning: Paper Code

CVPR 2022

Moslem Yazdanpanah*, Aamer Abdul Rahman*, Muawiz Chaudhary, Christian Desrosiers, Mohammad Havaei, Eugene Belilovsky, Samira Ebrahimi Kahou

AWARDS AND ACHIEVEMENTS

Master's Research Award: Received award worth \$64,000 for pursuing a research masters at Mila.

Paper Spotlight: Spotlight paper at the NeurIPS 2023 workshop on Goal-Conditioned RL. (Top 15%)

Amii Talent Bursary: Received award worth \$1500 to attend UpperBound 2024.

ICVSS 2024: Accepted for the 2024 International Computer Vision Summer School. (24% acceptance rate)

Tuition Fees Exception: Quebec exemption from international master's tuition fees.

TECHNICAL SKILLS

Programming: Python, R, C++ , MATLAB, Octave, SQL

Frameworks & Tools: PyTorch, TensorFlow, Pandas, HuggingFace, Matplotlib, Flask, Scikit-Learn, AWS, Azure, Git, LaTeX

Relevant Coursework: Representation Learning, Computer Vision, Reinforcement Learning, Applied Machine Learning, Robotics, Foundations of Data Science

Mathematics: Linear Algebra, Multivariate Calculus, Probability and Statistics

Languages: English, French, Malayalam